STPM 2023 PHYSICS 960/2 (U2)

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $N_A = 6.02 \times 10^{23}$ mol^{-1}, $\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1}, $k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9$ N m^2 C^{-2}.

Section A [15 marks]

1. A spherical conductor is surrounded by a Gaussian surface. If the area of the sphere is 3.0 m^2 and the surface charge density is +20 μC m^{-2}, what is the electric flux that passes through the surface?

A 7.5×10^5 N m^2 C^{-1}
B 2.3×10^6 N m^2 C^{-1}
C 6.8×10^6 N m^2 C^{-1}
D 2.6×10^8 N m^2 C^{-1}

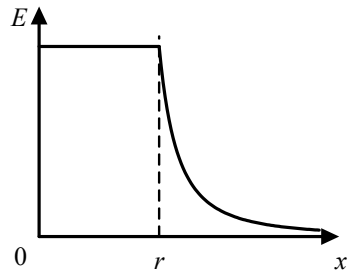
Answer: C

$$Q = \sigma A = (20 \times 10^{-6})(3.0) = 6.0 \times 10^{-5} \text{ C}$$
$$\Phi = \frac{Q}{\varepsilon_0} = \frac{6.0 \times 10^{-5}}{8.85 \times 10^{-12}} = 6.8 \times 10^6 \text{ N m}^2 \text{ C}^{-1}$$

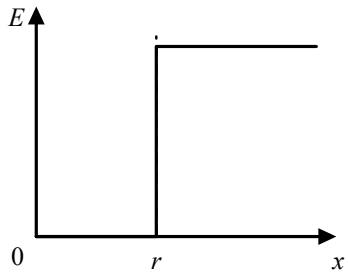
Explanation: Gauss's law: the flux depends only on the charge enclosed.

2. Which graph shows the relationship between electric field strength, E, and distance, x, for a positively charged hollow conducting sphere of radius r?

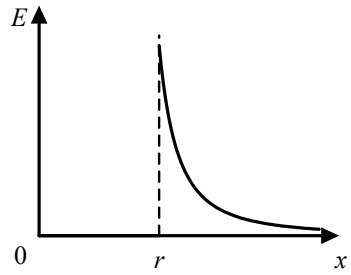
A



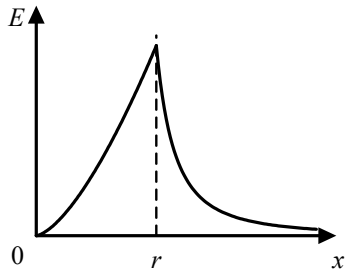
B



C



D



Answer: C

Inside a conducting sphere $E = 0$; just outside it is maximum, then it falls as $1/x^2$ (C).

Explanation: All the charge sits on the outer surface of a conductor.

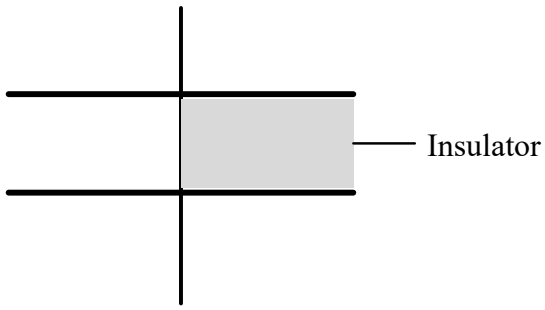
3. Which statement is **not** true about the capacitance of a capacitor?

A Capacitance is the ability of a capacitor to store electric charge in the plates.
B Capacitance is the ratio of the charge stored in the plates to the potential difference between the plates.
C The capacitance of a capacitor depends on the physical dimensions of the capacitor and the material between the plates.
D The capacitance of a capacitor depends on the charge stored in the capacitor and the potential difference between the plates.

Answer: D
Capacitance depends only on the dimensions and the dielectric. Statement D (depends on Q and V) is not true.

Explanation: $C = Q/V$ is a ratio: doubling V doubles Q.

4. A parallel plate capacitor that is half filled with an insulator between the plates is shown in the diagram below.



The dielectric constant of the insulator is 3.4. If the area of the plates is 8.4×10^{-5} m^2 and the distance between the plates is 1.5×10^{-4} m, what is the capacitance of the capacitor?

A 1.9 pF B 2.5 pF C 8.4 pF D 11 pF

Answer: D
The insulator fills half of the plate area, so the capacitor acts as two capacitors of area $A/2$ in parallel.

$$C = \frac{\varepsilon_0(A/2)}{d}(1 + 3.4) = \frac{(8.85 \times 10^{-12})(4.2 \times 10^{-5})}{1.5 \times 10^{-4}}(4.4) = 1.1 \times 10^{-11}\ \text{F} = 11\ \text{pF}$$

Explanation: Side-by-side regions are in parallel; layered regions would be in series.

5. Two wires, X and Y, have the same current density. If the free-electron density of X is twice the free-electron density of Y, the drift velocity of the electrons in X is

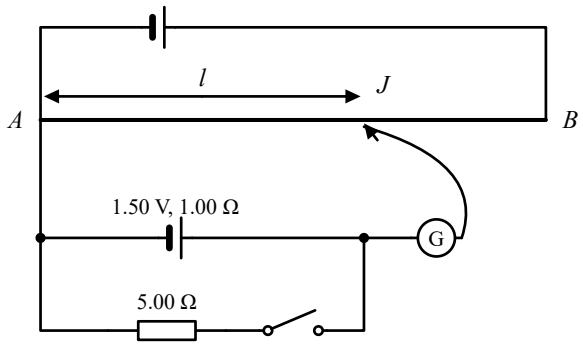
A half the drift velocity of the electrons in Y
B twice the drift velocity of the electrons in Y
C four times the drift velocity of the electrons in Y
D one quarter the drift velocity of the electrons in Y

Answer: A

$$J = nev_d;\ J \text{ same} \Longrightarrow v_d \propto \frac{1}{n}\text{: half the drift velocity in } Y$$

Explanation: More carriers share the same current density.

6. A dry cell with e.m.f. 1.50 V and internal resistance 1.00 Ω is connected across a potentiometer and is connected with a resistor of resistance 5.00 Ω and a switch as shown in the circuit diagram below.



If the balanced length, l, is 75.0 cm when the switch is opened, what is the balanced length when the switch is closed?

A 12.5 cm B 37.5 cm C 62.5 cm D 75.0 cm

Answer: C

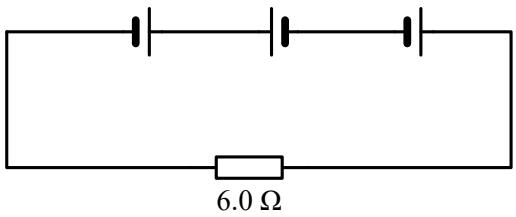
$$\text{Gradient} = \frac{1.50}{75.0} = 0.0200 \text{ V cm}^{-1}$$

$$\text{Switch closed: } V = 1.50 \times \frac{5.00}{5.00 + 1.00} = 1.25 \text{ V}$$

$$l = \frac{1.25}{0.0200} = 62.5 \text{ cm}$$

Explanation: With the switch open no current flows in the cell at balance, so the length measures the e.m.f.

7. Three dry cells are connected to the terminals as shown in the circuit diagram below.



Each dry cell has e.m.f. 2.0 V and internal resistance 0.50 Ω. What is the power dissipated in the resistor of resistance 6.0 Ω?

A 0.43 W B 0.57 W C 0.67 W D 3.8 W

Answer: A

The middle cell is reversed, so the net e.m.f. = 2.0 + 2.0 − 2.0 = 2.0 V; total internal resistance = 1.5 Ω.

$$I = \frac{2.0}{6.0 + 1.5} = 0.267 \text{ A}; \; P = I^2R = (0.267)^2(6.0) = 0.43 \text{ W}$$

Explanation: A reversed cell subtracts its e.m.f. but still adds its internal resistance.

8. A particle of mass m and charge q enters perpendicularly into a uniform magnetic field B with a speed v. The radius of the circular path of the particle is

A directly proportional to m
B directly proportional to q
C inversely proportional to v
D directly proportional to B

Answer: A

$$r = \frac{mv}{qB}: \text{ directly proportional to } m$$

Explanation: r is inversely proportional to q and B, and directly proportional to v.

9. An electron moves in a uniform magnetic field and uniform electric field which is perpendicular to each other. If the magnetic field is removed, the electron will move with

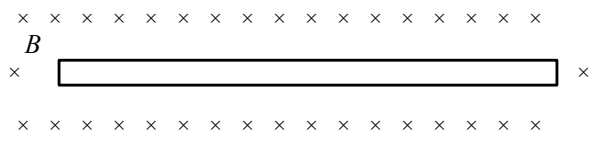
A the same speed in a circle
B the same speed in a parabola
C the same speed in a straight path
D the increasing speed in a straight path

Answer: B

The electron was undeflected, so its velocity is perpendicular to **E**. With **B** removed only the constant electric force acts, perpendicular to the initial velocity: the path is a parabola (B).

Explanation: Strictly, the speed then increases; the velocity component perpendicular to **E** is what stays the same. B is the intended answer.

10. A straight wire of length 10 cm that carries a current of 20 A to the east in a magnetic field, B, of 4.0 T directed into the page is shown in the diagram below.



What are the magnitude and direction of the magnetic force that acts on the wire?

A 0.020 N to the north
B 0.020 N to the south
C 8.0 N to the north
D 8.0 N to the south

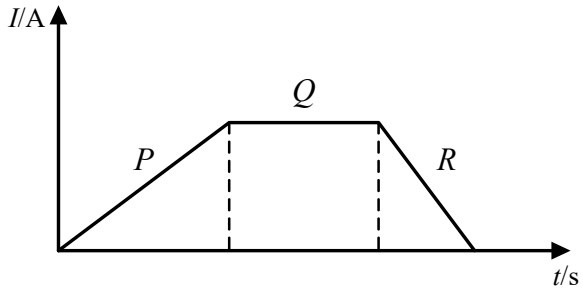
Answer: C

$$F = BIL = (4.0)(20)(0.10) = 8.0 \text{ N}$$

Fleming's left-hand rule (current east, field into the page): force to the north.

Explanation: Use the length in metres.

11. The variation of current, I, that flows through a pure inductor with time, t, is shown in the graph below.



Which is the correct arrangement of induced e.m.f. at P, Q and R intervals in increasing order of magnitude?

A $P < Q < R$ B $Q < R < P$ C $Q < P < R$ D $R < Q < P$

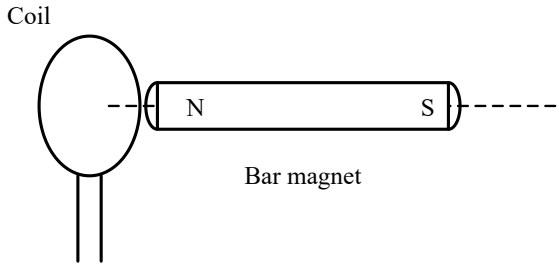
Answer: C

$$|\varepsilon| = L\frac{dI}{dt}\text{: proportional to the gradient of the graph}$$

Q has zero gradient, P a moderate gradient and R the steepest: $Q < P < R$.

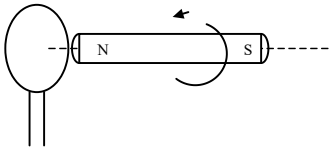
Explanation: A constant current induces no e.m.f.

12. A bar magnet placed along the axis of a coil is shown in the diagram below.

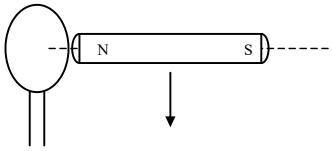


Which movement does **not** produce any induced e.m.f. at both terminals of the coil when the magnet is moved in the direction of the arrow?

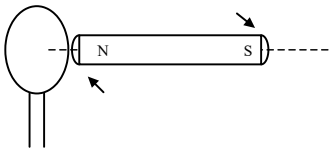
A



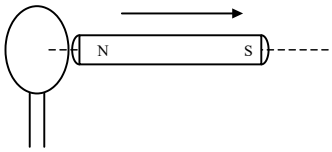
B



C



D

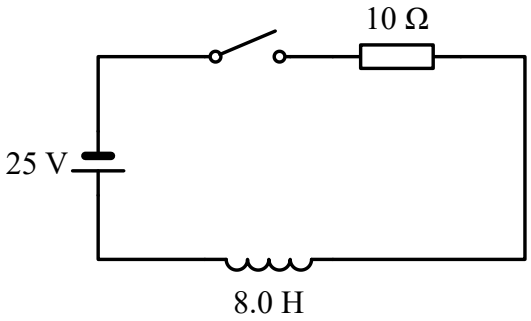


Answer: A

Rotating the magnet about its own axis (A) does not change the flux through the coil, so no e.m.f. is induced.

Explanation: The field of a bar magnet is symmetric about its axis.

13. A resistor of resistance 10 Ω and an inductor of self inductance 8.0 H are connected in series with a battery of e.m.f. 25 V as shown in the circuit diagram below.



What is the energy stored in the inductor and the back e.m.f. when a steady current is achieved?

	Energy stored	*Back e.m.f.*
A	25 J	25 V
B	25 J	0 V
C	80 J	25 V
D	80 J	0 V

Answer: B

$$\text{Steady current } I = \frac{25}{10} = 2.5 \text{ A}; \; U = \frac{1}{2}LI^2 = \frac{1}{2}(8.0)(2.5)^2 = 25 \text{ J}$$

The current is constant, so $dI/dt = 0$ and the back e.m.f. is 0 V.

Explanation: The inductor affects only how fast the current grows.

14. A sinusoidal current with peak current I_0 flows through a resistor of resistance R. What is the mean power dissipated in the resistor?

A $\frac{I_0^2R}{4}$ B $\frac{I_0^2R}{2}$ C $\frac{I_0^2R}{\sqrt{2}}$ D I_0^2R

Answer: B

$$P = I_{\text{rms}}^2R = (\frac{I_0}{\sqrt{2}})^2R = \frac{I_0^2R}{2}$$

Explanation: The mean of $\sin^2$ over a cycle is ½.

15. A parallel plate capacitor of reactance 340 Ω is connected to a sinusoidal voltage supply. The voltage supply produces 20 V (r.m.s.) at angular frequency 90π rad s^{-1}. What is the instantaneous maximum charge stored in the capacitor?

A 4.2×10^{-4} C B 2.9×10^{-4} C C 2.1×10^{-4} C D 1.5×10^{-4} C

Answer: B

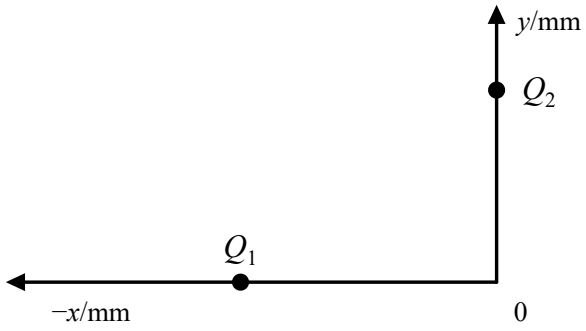
$$C = \frac{1}{\omega X_C} = \frac{1}{(90\pi)(340)} = 1.04 \times 10^{-5}\ \text{F}$$

$$Q_0 = CV_0 = (1.04 \times 10^{-5})(20\sqrt{2}) = 2.9 \times 10^{-4}\ \text{C}$$

Explanation: Use the peak voltage, not the r.m.s. value.

Section B [15 marks]

16. Two charges, Q_1 and Q_2, are placed in the xy-plane at positions (−3, 0) mm and (0, 2) mm as shown in the diagram below.



The charges of Q_1 and Q_2 are +30 μC and −50 μC respectively.

(a) Calculate the resultant electric field at origin. [3]

Answer

$E_1 = \dfrac{(9.0 \times 10^9)(30 \times 10^{-6})}{(3.0 \times 10^{-3})^2} = 3.0 \times 10^{10}$ N C^{-1} along $+x$ (away from Q_1) — 1

$E_2 = \dfrac{(9.0 \times 10^9)(50 \times 10^{-6})}{(2.0 \times 10^{-3})^2} = 1.125 \times 10^{11}$ N C^{-1} along $+y$ (towards Q_2) — 1

$E = \sqrt{((3.0 \times 10^{10})^2 + (1.125 \times 10^{11})^2)} = 1.16 \times 10^{11}$ N C^{-1}, at 75° above the $+x$-axis — 1

3

Explanation: The two fields are perpendicular, so use Pythagoras.

(b) Calculate the net electric potential at origin. [3]

Answer

$V = k\left(\frac{Q_1}{r_1} + \frac{Q_2}{r_2}\right)$ 1

$V = (9.0 \times 10^9)\left(\frac{30 \times 10^{-6}}{3.0 \times 10^{-3}} - \frac{50 \times 10^{-6}}{2.0 \times 10^{-3}}\right)$ 1

$V = -1.35 \times 10^8$ V 1

3

Explanation: Potential is a scalar: keep the signs of the charges.

(c) Determine the new position of Q_2 if electric potential is zero at origin. [2]

Answer

$\frac{30}{3.0} = \frac{50}{r} \Longrightarrow r = 5.0$ mm 1

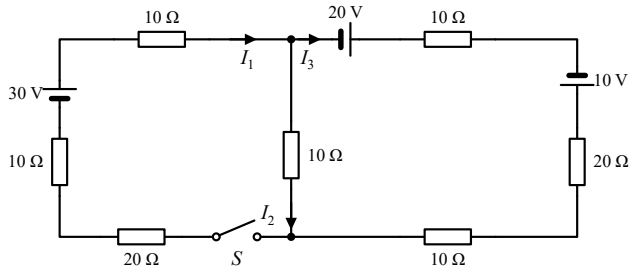
Q_2 at (0, 5.0) mm 1

2

Total for question **8**

Explanation: Any point 5.0 mm from the origin works; keeping it on the y-axis gives (0, 5.0) mm.

17. Several e.m.f.s, resistors and a switch, S, are connected as shown in the circuit diagram below.



I_1, I_2 and I_3 show the current flow in the circuit. Determine the value of I_2

(a) when S is opened [4]

Answer

With S open no current flows in the left branch: $I_1 = 0$, so $I_3 = -I_2$ 1

Single loop (middle and right branches): net e.m.f. = 20 − 10 = 10 V 1

Total resistance = 10 + 10 + 20 + 10 = 50 Ω; current = $\frac{10}{50}$ = 0.20 A 1

This current flows up through the middle resistor, opposite to the arrow: $I_2 = -0.20$ A 1

4

Explanation: A negative answer means the current flows opposite to the direction assumed.

(b) when S is closed [3]

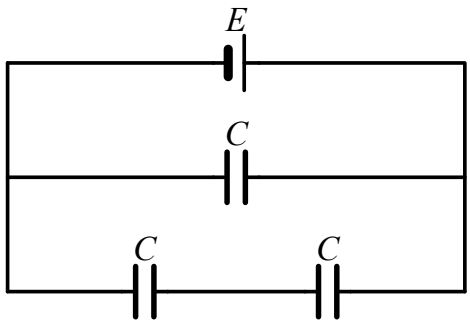
Answer

Junction: $I_1 = I_2 + I_3$	1
Left loop: $30 = 40I_1 + 10I_2$; right loop: $10 = 40I_3 - 10I_2$	1
Solving: $30 - 10I_2 = 40I_2 + 10 + 10I_2 \Longrightarrow I_2 = 0.33$ A	1
	3
Total for question	**7**

Explanation: Then $I_1 = 0.67$ A and $I_3 = 0.33$ A.

Section C [30 marks]

18. (a) Three identical capacitors are connected to an e.m.f., E, as shown in the circuit diagram below.



(i) Derive an expression for the effective capacitance in terms of C. [3]

Answer

Bottom branch, two in series: $\frac{1}{C_s} = \frac{1}{C} + \frac{1}{C} \Longrightarrow C_s = \frac{C}{2}$	1
In parallel with C: $C_{eff} = C + \frac{C}{2}$	1
$C_{eff} = 1.5C$	1
	3

Explanation: Series: same charge; parallel: same p.d.

(ii) Determine the total energy stored in terms of C and E. [2]

Answer

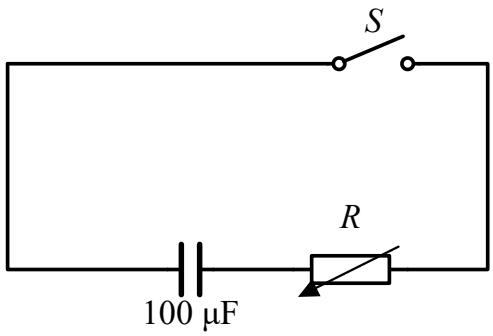
$U = \frac{1}{2}C_{\text{eff}}E^2 = \frac{1}{2}(1.5C)E^2$ 1

$U = \frac{3}{4}CE^2$ 1

2

Explanation: The full e.m.f. appears across the combination.

(b) A fully charged parallel plate capacitor of capacitance 100 μF, a variable resistor, R, and a switch, S, are connected as shown in the circuit diagram below.



(i) When S is closed, the charge reduces to half in time of 3.0 ms. Determine the resistance of R. [4]

Answer

$Q = Q_0 e^{-t/RC}$ 1

$\frac{1}{2} = e^{-t/RC} \Rightarrow t = RC \ln 2$ 1

$R = \frac{t}{C \ln 2} = \frac{3.0 \times 10^{-3}}{(100 \times 10^{-6}) \ln 2}$ 1

$R = 43.3\ \Omega$ 1

4

Explanation: The half-life of a discharge is $0.693RC$.

(ii) If a material of dielectric constant 4.2 is inserted in between the plates, determine the time constant. [2]

Answer

$C' = 4.2 \times 100\ \mu\text{F} = 420\ \mu\text{F}$; $\tau = RC' = (43.3)(420 \times 10^{-6})$ 1

$\tau = 0.0182$ s 1

2

Explanation: The dielectric multiplies the capacitance by ε_r.

(c) Describe the effect of dielectric on the capacitance when a capacitor is connected to a potential difference. [4]

Answer

The p.d. across the plates is kept constant by the supply	1
The dielectric is polarised; its induced surface charges reduce the field between the plates	1
More charge flows from the supply onto the plates to keep V constant	1
$C = \frac{Q}{V}$ increases (by the factor ε_r)	1
	4
Total for question	**15**

Explanation: The stored energy $\frac{1}{2}CV^2$ also increases.

19. (a) Explain the motion of free electrons when a potential difference is applied across a metal conductor. [4]

Answer

Free electrons normally move randomly at high speed, with no net flow	1
The p.d. sets up an electric field in the conductor	1
The electrons are accelerated opposite to the field between collisions with the lattice ions	1
They gain a small net drift velocity opposite to the field, giving a current	1
	4

Explanation: The collisions transfer energy to the lattice (heating).

(b) A constantan wire of length 100 cm and diameter 0.200 mm is used in a potentiometer. If the resistivity of constantan is 4.90×10^{-7} Ω m, determine the resistance of the wire. [3]

Answer

$A = \pi(0.100 \times 10^{-3})^2 = 3.14 \times 10^{-8}$ m^2	1
$R = \frac{\rho l}{A} = \frac{(4.90 \times 10^{-7})(1.00)}{3.14 \times 10^{-8}}$	1
$R = 15.6$ Ω	1
	3

Explanation: Use the radius, half of 0.200 mm.

(c) A copper wire of radius 0.850 mm has a density of 8.96×10^3 kg m^{-3}. Assume that each atom contributes one electron and the current density is uniform.

[Molar mass of copper is 63.54×10^{-3} kg mol^{-1}.]

(i) Calculate the number of electrons per unit volume of the wire. [4]

Answer

Number of moles per m^3 $= \frac{\rho}{M} = \frac{8.96 \times 10^3}{63.54 \times 10^{-3}} = 1.41 \times 10^5$ mol m^{-3} 1

Atoms per m^3 $= (1.41 \times 10^5)(6.02 \times 10^{23})$ 1

One free electron per atom 1

$n = 8.49 \times 10^{28}$ m^{-3} 1

4

Explanation: $n = \rho N_A/M$.

(ii) Calculate the drift velocity of the electrons when the current flows are 20.0 mA. [2]

Answer

$v_d = \frac{I}{nAe} = \frac{20.0 \times 10^{-3}}{(8.49 \times 10^{28})\pi(0.850 \times 10^{-3})^2(1.60 \times 10^{-19})}$ 1

$v_d = 6.5 \times 10^{-7}$ m s^{-1} 1

2

Explanation: Drift velocities are very small.

(iii) State the changes in current density if the copper wire is heated. Give your reason to the changes in the current density. [2]

Answer

The current density decreases (for the same p.d.) 1

Heating makes the ions vibrate more, so collisions increase and the resistance increases, reducing the current 1

2

Total for question **15**

Explanation: $J = \sigma E$, and the conductivity of a metal falls as it gets hotter.

20. (a) A capacitor of capacitance, C, is connected to an alternating source of frequency, f. If the current, $I = I_0 \sin 2\pi ft$, derive an expression for the reactance of the capacitor in terms of C and f. [4]

Answer

$Q = CV \Longrightarrow I = \frac{dQ}{dt} = C\frac{dV}{dt}$ 1

$V = \frac{1}{C}\int I_0 \sin \omega t \, dt = -\frac{I_0}{\omega C} \cos \omega t$ 1

Peak voltage $V_0 = \frac{I_0}{\omega C}$ 1

$X_C = \frac{V_0}{I_0} = \frac{1}{\omega C} = \frac{1}{2\pi fC}$ 1

4

Explanation: The voltage across a capacitor lags the current by 90°.

(b) A capacitor of capacitance 50.0 μF is connected to a power supply of peak voltage 12.0 V and frequency 150 Hz. Determine

(i) the reactance of the capacitor in the circuit [1]

Answer

$X_C = \frac{1}{2\pi(150)(50.0 \times 10^{-6})} = 21.2\ \Omega$ 1

1

Explanation: Reactance falls as the frequency rises.

(ii) the r.m.s. current in the circuit [2]

Answer

$V_{rms} = \frac{12.0}{\sqrt{2}} = 8.49$ V; $I_{rms} = \frac{8.49}{21.2}$ 1

$I_{rms} = 0.400$ A 1

2

Explanation: The supply's peak voltage must be converted to r.m.s.

(c) In an a.c. circuit of a pure inductor, show that the current lags the voltage. [3]

Answer

$V = L\frac{dI}{dt}$ with $I = I_0 \sin \omega t$ 1

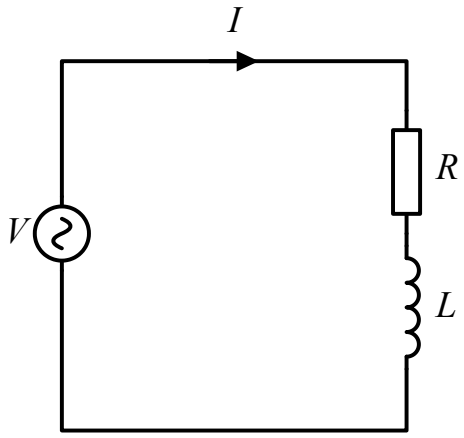
$V = \omega L I_0 \cos \omega t = V_0 \sin(\omega t + \frac{\pi}{2})$ 1

The voltage leads the current by $\frac{\pi}{2}$, i.e. the current lags the voltage by 90° 1

3

Explanation: The voltage is maximum when the current is changing fastest (passing through zero).

(d) A resistor and an inductor are connected in series to an a.c. supply as shown in the circuit diagram below.



The resistance of resistor, R, is 50 Ω and the inductance of inductor, L, is 30 mH. If the a.c. supply, $V = 240 \sin 100\pi t$, calculate the

(i) impedance of the circuit [3]

Answer

$X_L = \omega L = (100\pi)(30 \times 10^{-3}) = 9.42\ \Omega$ 1

$Z = \sqrt{(R^2 + X_L^2)} = \sqrt{(50^2 + 9.42^2)}$ 1

$Z = 50.9\ \Omega$ 1

3

Explanation: $\omega = 100\pi$ rad s^{-1} is read from the supply equation.

(ii) peak current in the circuit [2]

Answer

$I_0 = \frac{V_0}{Z} = \frac{240}{50.9}$ 1

$I_0 = 4.72$ A 1

2

Total for question **15**

Explanation: Peak voltage over impedance gives the peak current.